1. (2 points) For each symbol below, write its name or meaning in English. The first is filled in for you as an example.

(a) $\mathscr{P}$: power set = set of all subsets of a set

(b) $\mathbb{R}$: real numbers $\mathbb{Z}$ $\mathbb{N}$ $\mathbb{Q}$

(c) $\cap$: intersection

(d) $\cup$: union

(e) $|\cdot|$: cardinality, size, # elts

$\mathscr{P}(\{1,2,3\})$

$= \{\emptyset, \{1\}, \{1,3\}, \ldots\}$

2. (2 points) Circle True or False for each of the following statements.

$\{a\} \in \{\{a\}, b\}$

(a) $a \in \{\{a\}, b\}$: True or False

(b) $\{1,2\} \subseteq \{1,2,3,5\}$: True or False

(c) $11 \in \{x \in \mathbb{Z} : x < 10\}$: True or False

a in [[a], b] ?

[a] in

3. (3 points) Given the following preference lists as input for the hospital/medical student matching problem.

$$\begin{array}{llll} H_1: & S_3 & S_1 & S_2 \\ H_2: & S_3 & S_2 & S_1 \\ H_3: & S_2 & S_3 & S_1 \end{array} \qquad \begin{array}{llll} S_1: & H_1 & H_2 & H_3 \\ S_2: & H_3 & H_2 & H_1 \\ S_3: & H_1 & H_2 & H_3 \end{array}$$

(a) Is the matching $M = \{(H_1, S3), (H_2, S_1), (H_3, S_2)\}$ stable? Yes or No

(b) How do you know?

no unstable pairs

(h,s) not in M

$\notin M$

4. (3 points) The next question asks about the same preference lists, which are copied for your convenience.

$$\begin{array}{llll} H_1: & S_3 & S_1 & S_2 \\ H_2: & S_3 & S_2 & S_1 \\ H_3: & S_2 & S_3 & S_1 \end{array} \qquad \begin{array}{llll} S_1: & H_1 & H_2 & H_3 \\ S_2: & H_3 & H_2 & H_1 \\ S_3: & H_1 & H_2 & H_3 \end{array}$$

(a) Is the matching $M = \{(H_1, S_1), (H2, S_3), (H_3, S_2))\}$ stable? Yes or No

(b) How do you know?

there is an unstable pair

Given an input of hosp / students and pref lists, how to find a stable matching (if one exists)?

brute force?

all possible M $n!$

GALE–SHAPLEY (*preference lists for hospitals and students*)

INITIALIZE M to empty matching.

WHILE (some hospital h is unmatched and hasn't proposed to every student)

$s \leftarrow$ first student on h's list to whom h has not yet proposed.

IF (s is unmatched)

Add h–s to matching M.

ELSE IF (s prefers h to current partner h')

Replace h'–s with h–s in matching M.

ELSE

s rejects h.

RETURN stable matching M.

1. Once a student becomes matched, they never become unmatched. T
2. Once a student becomes matched, they only trade up for better hospitals. T
3. Once a student becomes matched, they only trade down for worse hospitals. F
4. Once a hospital becomes matched, it never becomes unmatched. F
5. A hospital only trades up for better students. F
6. A hospital only trades down for worse students. T
7. A hospital will never propose to the same student twice. T
8. The last student a hospital proposes to <u>is</u> the one it ends up with. T

GALE–SHAPLEY (*preference lists for hospitals and students*)

INITIALIZE M to empty matching.

WHILE (some hospital h is unmatched and hasn't proposed to every student)

  $s \leftarrow$ first student on h's list to whom h has not yet proposed.

  IF (s is unmatched)

    Add h–s to matching M.

  ELSE IF (s prefers h to current partner h')

    Replace h'–s with h–s in matching M.

  ELSE

    s rejects h.

RETURN [redacted]g M.

<u>Claim</u> the while executes at most n^2 times.

Pf by 7

Claim the set M returned by G-S is a stable matching